

Experiment No 3 Hybrid Cloud

Setting Up Hybrid Connectivity Experiment with setting up a secure VPN connection between an on-premises data center and a public cloud provider such as AWS or Azure.

Configure a hybrid network topology using virtual private clouds (VPCs), subnets, and route tables.

Steps:

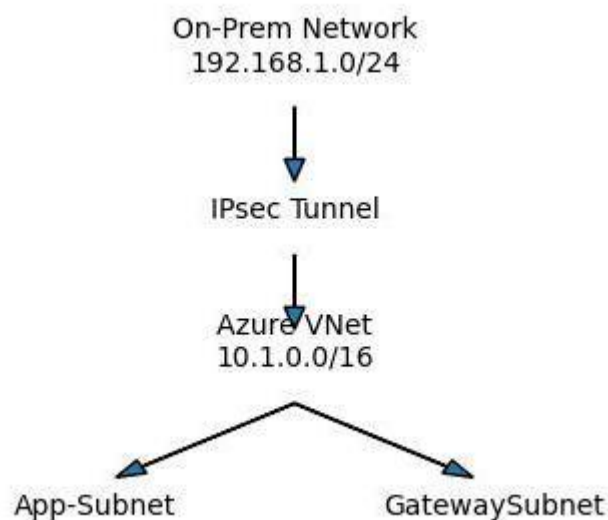
Create a Virtual Network (VNet)

Create Subnets

Create a VPN Gateway

Simulate an “On-Prem” network

Connect them using Site-to-Site VPN



STEP 1: Create a Resource Group

Think of Resource Group as a folder.

1. Login to Azure Portal
2. In search bar type **Resource groups**
3. Click → **Create**
4. Fill:

- Subscription → Your student subscription ○ Resource group name → Hybrid-RG
- Region → Choose same region everywhere (example: Central India)

5. Click **Review + Create**

6. Click **Create**

STEP 2: Create Virtual Network (VNet)

VNet = Your cloud network.

1. Search **Virtual Networks**
2. Click **Create**
3. Fill:

Basics Tab

- Resource Group → Hybrid-RG
- Name → Cloud-VNet
- Region → SAME region as resource group

Click **Next**

IP Addresses Tab

- Address Space:

10.1.0.0/16

Remove default subnet.

Click + **Add Subnet**

Create Subnet 1: •

Name: App-Subnet

- Address range:

10.1.1.0/24

Click **Add**

IMPORTANT: Add Gateway Subnet

Click + **Add Subnet** again:

- Name: GatewaySubnet (must be EXACT name)
- Address range:

10.1.255.0/27

Click Add

Now click **Review + Create**

Click **Create**

VNet created

STEP 3: Create Virtual Network Gateway (VPN Gateway)

This is Azure side VPN device.

1. Search → **Virtual network gateways**
2. Click → Create

Fill details:

Basics:

- Resource Group → Hybrid-RG
- Name → Azure-VPN-Gateway
- Region → Same region
- Gateway Type → VPN
- ☐ VPN Type → Route-based
- ☐ SKU → Select VpnGw1 (basic supported)
- ☐ Generation → Generation1

Virtual Network Section

Select:

Cloud-VNet

Public IP Address

Click **Create new** Name
it:

VPN-Gateway-IP

Click OK

Click **Review + Create** Click
Create

⚠ IMPORTANT:

This takes **30–45 minutes** to deploy. Be patient.

STEP 4: Simulate On-Prem Network Since you
don't have real data center, we simulate.

We create:

Local Network Gateway (represents on-prem)

1. Search → **Local Network Gateways**
2. Click → Create

Fill:

- Resource Group → Hybrid-RG
- Name → OnPrem-Gateway
- Region → Same region

IP Address:

Put any dummy public IP (example):

1.1.1.1

Address Space:

Add:

192.168.1.0/24

Click **Review + Create**
Click **Create**

Done

Create VPN Connection

Now connect Azure to On-Prem.

1. Go to **Virtual Network Gateways**
2. Click your gateway → Azure-VPN-Gateway
3. On left side click **Connections**
4. Click + **Add**

Fill:

- Connection Name → Hybrid-Connection
- Connection Type → Site-to-Site (IPsec)
- Virtual Network Gateway → Auto selected
- ☐ Local Network Gateway → OnPrem-Gateway
- ☐ Shared Key → Type simple password:

Azure123 Click

OK

VPN connection created

STEP 6: Verify Connection

1. Open Azure-VPN-Gateway
2. Click **Connections**
3. Check Status

If configured with real on-prem router → it will show Connected.

Since we used dummy IP, it will show Not Connected.

OUTPUT:

Create connection

Basics Settings Tags Review + create

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute.

[Learn more about VPN Gateway](#)

[Learn more about ExpressRoute](#)

Project details

Subscription *	Azure for Students
Resource group *	Hybrid

[Create new](#)

Instance details

Connection type *	VNet-to-VNet
Establish bidirectional connectivity	☐
Name *	
Region *	Central India

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *	Azure for Students
Resource group name *	Hybrid
Region *	(Asia Pacific) Central India

Microsoft Azure

Search resources, services, and docs (G+)

Home > Hybrid connectivity | ExpressRoute gateways

Create virtual network gateway

BasicsTagsReview + create

Azure has provided a planning and design guide to help you configure the various VPN gateway options. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
Azure for Students

Resource group ⓘ
Select a virtual network to get resource group

Instance details

Name *
Azure-VPN-Gateway

Region *
Central India
[Deploy to an Azure Extended Zone](#)

Gateway Configuration

Gateway type ⓘ
☒ VPN ☐ ExpressRoute

SKU * ⓘ
VpnGw2AZ

Generation ⓘ
Generation2

Enable Advanced Connectivity ⓘ
☐ Enabled ☒ Disabled

Virtual network * ⓘ

Review + createPreviousNext : Tags >Download a template for automation

Microsoft Azure

Search resources, services, and docs (G+)

Home > Network foundation | Virtual networks

Create virtual network

BasicsSecurityIP addressesTagsReview + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you want.

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. C virtual network address space into smaller ranges for use by your applications. When you deploy, assign the resource an IP address from the subnet. [Learn more](#)

☐ Allocate using IP address pools. [Learn more](#)

+ Add a subnet

Subnets	IP address range	Size	NAT
	10.1.0.0	/16	
	10.1.0.0 - 10.1.255.255	65,536 addresses	

Add IPv4 address space

PreviousNextReview + create

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ
Default

Name * ⓘ
App-Subnet

IPv4

Include an IPv4 address space ☒

IPv4 address range ⓘ
10.1.0.0/16
10.1.0.0 - 10.1.255.255

Starting address * ⓘ
10.1.0.0

Size ⓘ
/24 (256 addresses)

Subnet address range ⓘ
10.1.0.0 - 10.1.0.255

IPv6

Include an IPv6 address space ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Enable private subnet (no default outbound access) ☐

After March 31, 2026, private subnet will be the default selection for new virtual networks. [Learn more](#)

AddCancel

Give feedback

Virtual network gateway

Search

+ Add Refresh

Search connections

Name	Status	Connection type	Peer
Hybrid-Connection	Unknown	Site-to-site (IPsec)	OnPrem-Gateway

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Connections

Point-to-site configuration

Maintenance

Properties

Microsoft Azure

Search resources, services, and docs (G+)

Home > Hybrid connectivity > Local network gateways

Create local network gateway

Basics Advanced Review + create

A local network gateway is a specific object that represents an on-premises location (the site) for routing purposes. [Learn more](#)

Project details

Subscription * Azure for Students

Resource group * (New) Hybrid-RG

Create new

Instance details

Region * Central India

Name * OnPrem-Gateway

Endpoint IP address P2P

IP address * 1.1.1.1

Address Space(s)

192.168.1.0/24

Add additional address range

Review + create Previous Next: Advanced >